Raman Spectroscopy

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Raman spectroscopy with a 532 nm laser was used to distinguish carbon allotropes, test whether three diamond-like samples are diamond, estimate the thickness of a graphene film and identify an unlabelled semiconductor wafer. Bands were fitted in every run, and silicon and diamond reference lines fixed the wavenumber calibration within 1 cm^{-1}. The sp^2 samples share the G band near 1580 cm^{-1} but differ in disorder and stacking. Highly oriented pyrolytic graphite showed no D band ($I_D/I_G < 0.02$) and the two-component 2D band of AB-stacked graphite. A carbon rod gave $I_D/I_G = 0.13 \pm 0.02$ with a G band unexpectedly downshifted to 1566 ± 1 cm^{-1}, and a nanotube sample gave 1.24 ± 0.08. Diamond gave its sp^3 line at 1331.5 ± 1.0 cm^{-1}, although the peak saturated the detector. None of the diamond-like samples showed that line. One is amorphous carbon (G at 1536 ± 10 cm^{-1}, $I_D/I_G = 0.81 \pm 0.02$), one showed carbon bands on a strong photoluminescence background in one of four trials, and one showed no carbon bands. The graphene film is thicker than a monolayer (2D width 47 ± 4 cm^{-1}), but its layer number could not be fixed. The wafer is crystalline silicon (520.9 ± 1.0 cm^{-1}). The unknown laser power at the sample prevents comparison of absolute intensities.

1. INTRODUCTION

Most analytical techniques require at least some sample preparation, but Raman spectroscopy works on materials essentially as they are. A laser beam focused onto the sample scatters off the lattice vibrations, and the energy shift of the scattered photons encodes the vibrational frequencies of the material. Because crystal structure, bonding geometry and even the number of atomic layers all affect these frequencies, a single spectrum can reveal a sample's identity and structural quality.

This is particularly useful for carbon. Graphite, diamond, nanotubes and graphene contain only carbon atoms but differ in their bonding. sp^2 carbon (graphite, graphene, nanotubes) and sp^3 carbon (diamond) produce entirely different Raman spectra, and within the sp^2 family, disorder, stacking order and dimensionality introduce further differences. Raman spectroscopy therefore distinguishes materials that are identical by elemental analysis.

This experiment follows the four tasks of the laboratory manual [1]: identifying the vibrational bands of several carbon allotropes, testing whether three diamond-like samples are genuine diamond, estimating the layer count of a graphene film on SiO_2/Si, and identifying an unlabelled semiconductor wafer.

2. THEORY

2.1 Raman Scattering

A molecule has electronic levels separated by a few electron-volts, vibrational levels about 0.1 eV apart within each electronic state, and rotational levels about 10^{-4} eV apart (Fig. 1) [2,3]. In a solid the rotational levels are absent, and the vibrational levels become the quantised lattice vibrations, the phonons. Raman scattering changes the vibrational state by about 0.1 eV, which shifts the scattered light measurably from the 2.3 eV laser photons.

When monochromatic laser light hits a material, most photons scatter elastically, leaving with the energy they arrived with (Rayleigh scattering). A small fraction exchange energy with lattice vibrations (phonons) and scatter inelastically. A photon that gives energy to the lattice leaves with a longer wavelength (Stokes scattering); one that absorbs a thermally excited phonon leaves with more energy (anti-Stokes scattering) [1,4].

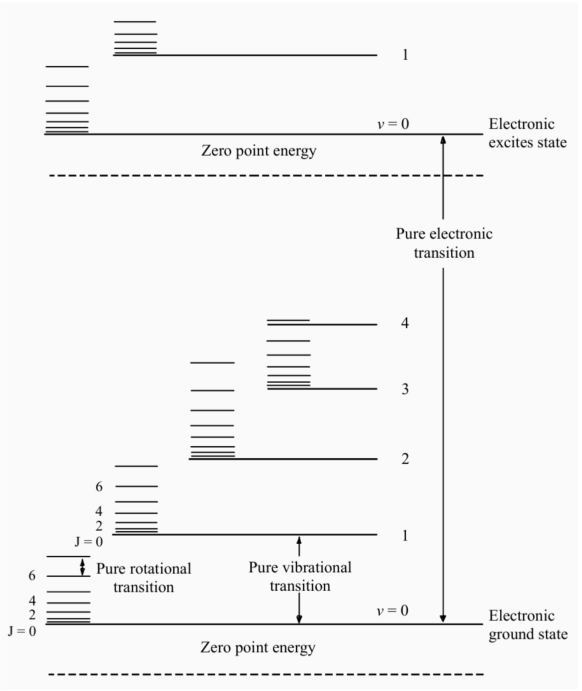
Fig. 2 shows the three processes. The Raman spectrum plots scattered intensity against the shift from the laser line, expressed as a wavenumber [4]:

$$\Delta\tilde{\nu} = \frac{1}{\lambda_0} - \frac{1}{\lambda_s}, \tag{1}$$

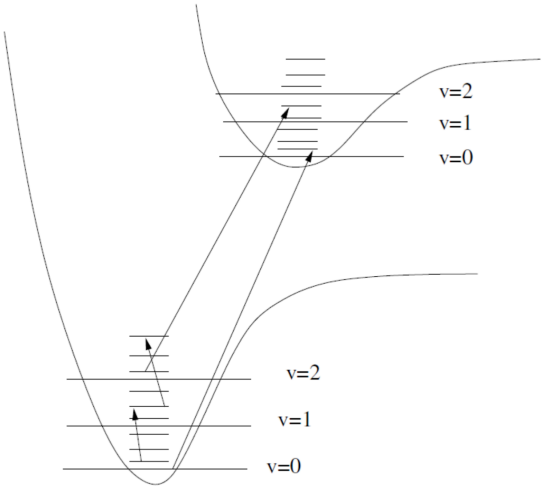
where λ_0 is the laser wavelength and λ_s the scattered wavelength; with both in centimetres, $\Delta\tilde{\nu}$ is in cm^{-1}. Each peak corresponds to a vibrational mode. At room temperature most of the material is in its vibrational ground state, so Stokes lines are much stronger than anti-Stokes lines and the Stokes spectrum is the one recorded [4]. A bond's vibrational frequency rises with its stiffness and falls with the masses involved, which is why the same element gives different spectra in different bonding geometries.

2.2 Carbon Allotrope Signatures

Carbon bonds as either sp^2 (planar, three-fold coordination) or sp^3 (tetrahedral, four-fold coordination). sp^2 carbon shows four main bands [5]. The G band near 1580 cm^{-1} is the in-plane C–C stretching mode, a zone-centre phonon present in all sp^2 carbon. The D band near 1350 cm^{-1} is a breathing mode of the carbon rings that becomes Raman-active only when defects, grain boundaries or edges break the translational symmetry. It arises from an intervalley double-resonance process and is dispersive,



(a)



(b)

FIG. 1. Energy levels of a diatomic molecule. (a) Electronic, vibrational (υ) and rotational (J) levels; the true electronic spacings are much larger, and the rotational spacings much smaller, than drawn. (b) Two electronic potential curves with their vibrational levels (long lines) and the rotational levels built on them (short lines). Reproduced from González Mena [2], after Ferraro, Nakamoto and Brown [3].

shifting with excitation wavelength, and a larger D/G intensity ratio indicates more disorder. A second defect-activated mode, D′ near 1620 cm⁻¹, comes from intravalley double resonance and appears as a shoulder on the high-frequency side of G. The 2D band near 2700 cm⁻¹, also called G′, is the overtone of the D mode. Its two phonons of opposite momentum conserve momentum, so it needs no defects, and its shape depends on the stacking of the layers, which makes it the standard thickness indicator for graphene.

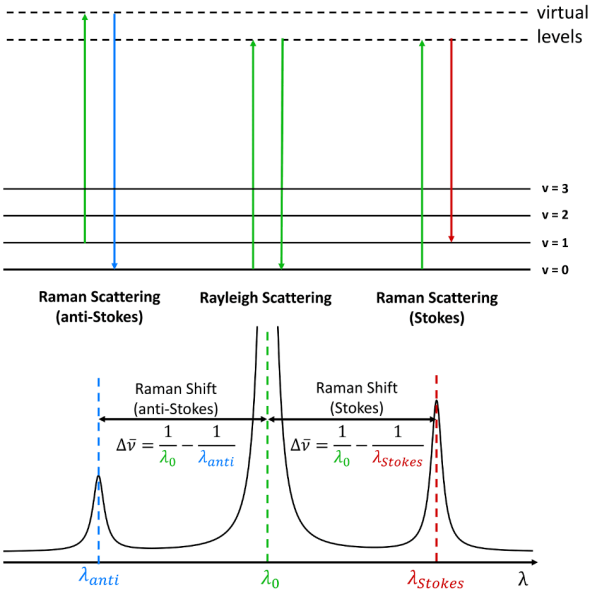


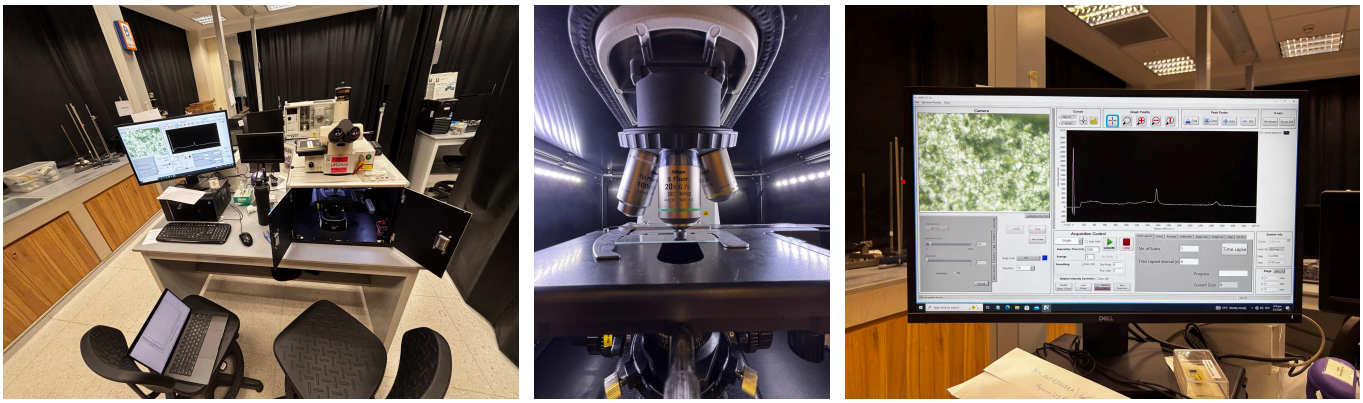
FIG. 2. Top: anti-Stokes, Rayleigh and Stokes scattering between vibrational levels $\upsilon = 0$–3 through short-lived virtual levels. Bottom: the resulting spectrum. The Rayleigh line sits at the laser wavelength λ_0; the Stokes line lies at longer wavelength and the anti-Stokes line at shorter wavelength, each shifted by the same wavenumber $\Delta\bar{\nu}$ of Eq. (1).

In sp³ carbon (diamond), the bonding supports a single sharp line near 1332 cm⁻¹ from the triply degenerate zone-centre optical phonon [4].

Diamond-like carbon (DLC) is an amorphous network of sp² and sp³ sites. Contrary to a common expectation, DLC does *not* show a shifted or broadened 1332 cm⁻¹ line: visible-light Raman spectra of DLC are dominated by the sp² sites, which produce broad D and G bands [6]. In the three-stage disorder model of Ferrari and Robertson [6], the G band moves from 1581 cm⁻¹ in graphite up to about 1600 cm⁻¹ in nanocrystalline graphite (stage 1), then down to about 1510 cm⁻¹ in amorphous carbon (stage 2) as the sp² clusters shrink and distort. Hydrogenated amorphous carbon (a-C:H) also shows a photoluminescence (PL) background that rises with hydrogen content [6].

2.3 Thickness Sensitivity in Layered Materials

The interlayer coupling of graphene layers splits the electronic bands, and the double-resonance process behind the 2D band samples these bands. In monolayer graphene the 2D band is a single sharp Lorentzian, more intense than G. In bilayer and few-layer graphene it broadens into several components and weakens relative to G. In bulk AB-stacked graphite it consists of two components, $2D_1$ and $2D_2$, with $2D_2$ (about 2720 cm⁻¹) the stronger [5].



(a) (b) (c)

FIG. 4. (a) The micro-Raman workstation: uRaman module (white housing) on a Nikon microscope, the sample enclosure (door open here, closed during measurements) and the acquisition computer. (b) The objective turret: Nikon S Fluor 20×/0.75 in the beam path, used for all measurements, between a TU Plan Fluor 100× and a lower-magnification objective (NA 0.30). (c) The acquisition software during the carbon-rod measurement: camera view of the rod surface (left), live spectrum (right) and the settings (1000 ms, 3 averages, smoothing with 6 side points and polynomial order 3).

3. METHODS

3.1 Instrument

All spectra were recorded on the micro-Raman uRaman-532TEC-Ci system of the teaching laboratory [1] (Fig. 3), with 532 nm excitation (532.1 nm measured), a Nikon microscope and a 20× objective (S Fluor, NA 0.75; Fig. 4). The laser runs at up to 100 mW. Its power was reduced first with a five-position attenuator knob (100% to 5% transmission) and then in the software [1]. The software reports 0 mW as the laser power, and the knob position was not noted, so the power at the sample is unknown. Before each sample measurement a dark spectrum was acquired at the same integration time and averaging, and the software subtracted it automatically [1]. Spectra were exported with 2048 points from −118 to 3431 cm^{-1}; the point spacing near 1300–1600 cm^{-1} is 1.7 cm^{-1}.

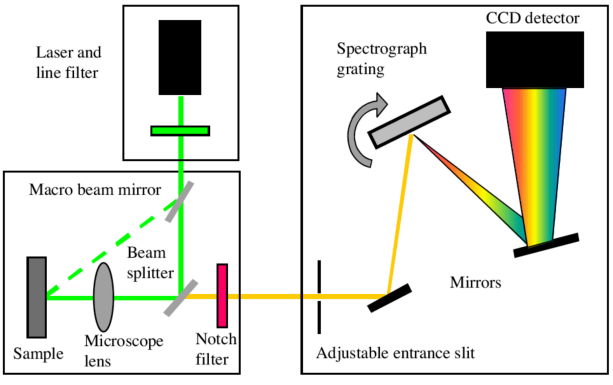


FIG. 3. Optical layout of a micro-Raman spectrometer. The laser passes a line filter and is directed by a beam splitter through the microscope lens onto the sample. The backscattered light returns through the beam splitter; a notch filter removes the Rayleigh line, and mirrors bring the remaining light through an adjustable entrance slit onto the spectrograph grating, which disperses it onto a CCD detector. A macro beam mirror allows samples outside the microscope to be measured.

3.2 Acquisition Parameters

The samples were a carbon rod (polycrystalline graphite), HOPG (highly oriented pyrolytic graphite), single-wall carbon nanotubes, a diamond, a graphene film on SiO_2/Si, three diamond-like samples and an unlabelled wafer (Fig. 5). Table I lists the acquisition settings.

TABLE I. Acquisition settings: number of runs, integration time, hardware averages and smoothing (the software's Savitzky–Golay filter, 6 side points, cubic). The graphene runs are five locations on the film. A fifth nanotube run (10 000 ms) was excluded (Sec. 4.1.3).

Sample	Runs	Time (ms)	Avg.	Smooth.
Carbon rod	3	1000	3	Yes
HOPG	3	1200	3	Yes
Diamond	3	1000	3	Yes
Unknown wafer	5	1000	1	Yes
Graphene	5	500	100	No
Nanotubes	4	1000–7000	10	Yes
DLC-I	4	700–2000	10	Yes
DLC-II	4	700	20	No
DLC-III	5	3000–7000	10	Yes

3.3 Analysis

Each run was divided by its integration time to give counts per second, so that runs at different integration times could be averaged; the figures show the run mean with the run-to-run standard deviation as a shaded band. Bands were fitted by least squares on a linear background: G, 2D and Si with pseudo-Voigt profiles (a weighted sum of a Lorentzian and a Gaussian of equal width), because their shape is partly set by the instrument, and D with a Lorentzian, fitted jointly with G over 1150–1720 cm^{-1}. The fit was applied to every run. A quoted value is the run mean. Its uncertainty is the run-to-run standard deviation or the fit standard error of the averaged spectrum, whichever is larger, combined for peak positions with a 1 cm^{-1} calibration uncertainty and widened for linewidths to cover the difference from a model-free half-maximum width (Sec. 4.1.6). Intensity ratios are ratios of fitted peak heights. Their uncertainty includes

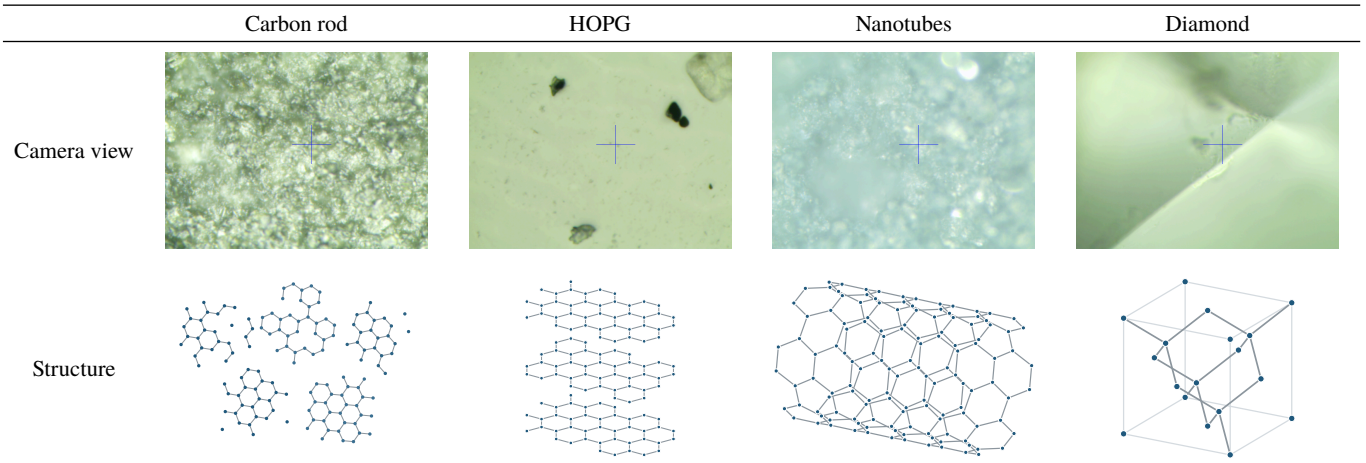


FIG. 5. The four allotrope samples. Top row: camera views through the 20× objective, with the cross marking the laser spot. Bottom row: sketches of their atomic structures. The carbon rod is polycrystalline graphite, small crystallites of hexagonal sheets at random orientations with defects and boundaries between them; HOPG consists of large, parallel, regularly stacked sheets; a single-wall nanotube is one sheet rolled into a cylinder; diamond is a tetrahedrally bonded lattice, shown as one cubic cell.

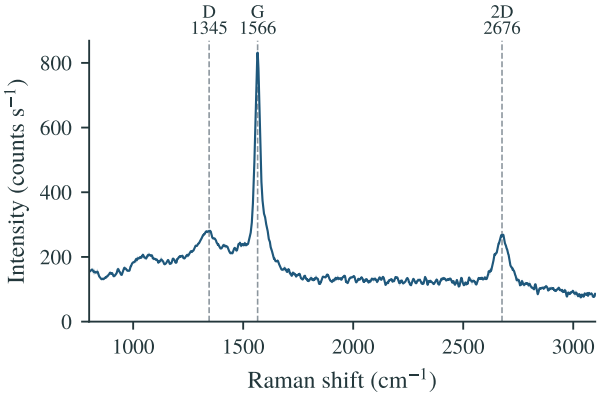


FIG. 6. Carbon rod: mean of 3 runs with the run-to-run standard deviation as the shaded band (narrower than the line). Dashed lines mark the fitted D, G and 2D positions.

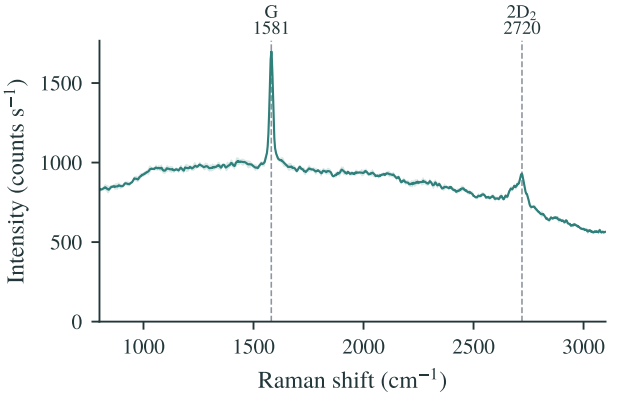


FIG. 7. HOPG: mean of 3 runs, drawn as in Fig. 6. No D band is visible near 1350 cm⁻¹; the 2D band has a low-frequency shoulder ($2D_1$) below the marked $2D_2$ peak.

the change when the linear background is replaced by a quadratic one, which was the largest analysis-choice effect found.

4. RESULTS AND ANALYSIS

4.1 Carbon Allotropes

4.1.1 Graphite (Carbon Rod)

The carbon rod (Fig. 6) shows the D, G and 2D bands of disordered graphite. The G band lies at 1566 ± 1 cm⁻¹ with a FWHM of 30 ± 3 cm⁻¹, the D band at 1345 ± 2 cm⁻¹ with $I_D/I_G = 0.13 \pm 0.02$, and the 2D band at 2676 ± 1 cm⁻¹. A D' shoulder at 1620 cm⁻¹ is resolved: including it in the fit gives it 8% of the G height and reduces the residual by 19%. The FWHM uncertainty reflects this shoulder, since a model-free half-maximum width (26 cm⁻¹) differs from the fitted one. The 2D band is a single broad peak (FWHM 74 ± 4 cm⁻¹) without the two-component shape of HOPG (Fig. 7), consistent with poorly stacked (turbostratic) graphite [5].

The G band is 15 cm⁻¹ below the HOPG value. This is not a calibration offset: HOPG, measured minutes later with the same calibration, agrees with graphite within 1 cm⁻¹. Nor is it the crystallite-size effect, which moves G *up* towards 1600 cm⁻¹ in nanocrystalline graphite [6]. The cause was not identified. Local laser heating, which softens the G mode, is a candidate that these data cannot test, since the power at the sample is unknown.

4.1.2 HOPG

HOPG (Fig. 7) shows the narrowest G band of the set, at 1581 ± 1 cm⁻¹ with a FWHM of 18.2 ± 0.5 cm⁻¹, close to the instrument limit of 12–16 cm⁻¹ (Sec. 4.1.6). No D band is detected: a D component fixed at 1350 cm⁻¹ has a height below twice its standard error, giving $I_D/I_G < 0.02$. The 2D band has the two-component shape of bulk AB-stacked graphite [5]: a fit with $2D_1$ at 2685 ± 13 cm⁻¹ and $2D_2$ at 2720 ± 2 cm⁻¹ lowers the residual by 38% relative to a single peak. A weak shoulder at 1620 cm⁻¹ (5% of G) is also present. Its assignment to D' is uncertain because D, normally the stronger defect band, is absent.

TABLE II. Fitted band positions, D/G peak-height ratios and G-band widths of the sp² samples. Position uncertainties combine the run-to-run scatter with the 1 cm⁻¹ calibration uncertainty. [a]$2D_2$ component; $2D_1$ at 2685 ± 13 cm⁻¹.

Sample	D (cm⁻¹)	G (cm⁻¹)	2D (cm⁻¹)	I_D/I_G	$FWHM_G$ (cm⁻¹)
Carbon rod	1345 ± 2	1566 ± 1	2676 ± 1	0.13 ± 0.02	30 ± 3
HOPG	not detected	1581 ± 1	2720 ± 2[a]	< 0.02	18.2 ± 0.5
Graphene	1350	1585 ± 2	2687.7 ± 1.5	0.04 ± 0.02	27 ± 2
Nanotubes	1337.7 ± 1.5	1576.1 ± 1.8	2675 ± 11	1.24 ± 0.08	68 ± 10
Reference [5]	1350	1580	2700		

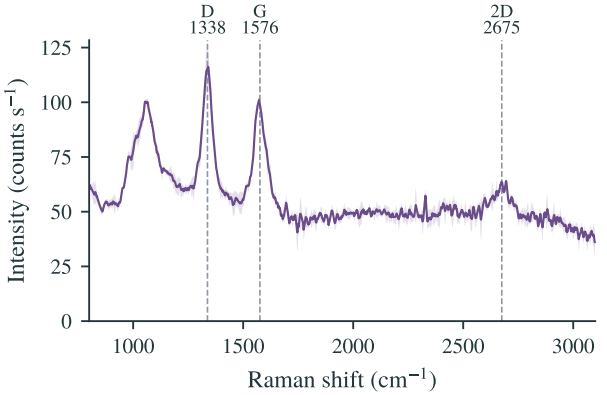


FIG. 8. Nanotube sample: mean of 4 runs (1000–7000 ms) in counts per second, drawn as in Fig. 6. The D band exceeds the G band.

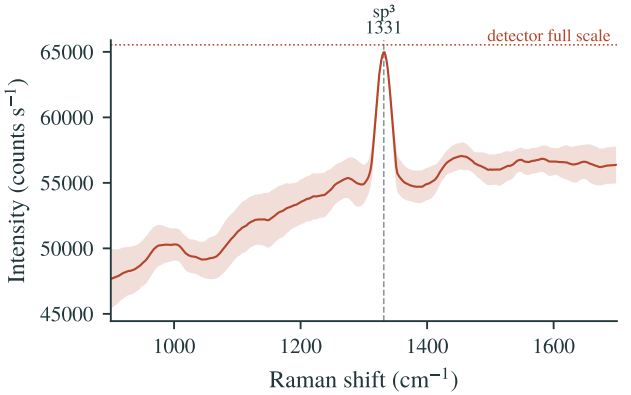


FIG. 9. Diamond: mean of 3 runs, drawn as in Fig. 6, on a strong photoluminescence background (offset intensity axis). The flattened top of the 1331 cm⁻¹ line reaches the detector full scale (dotted).

4.1.3 Carbon Nanotubes

The nanotube spectrum (Fig. 8) has a D band at 1337.7 ± 1.5 cm⁻¹ that is stronger than the G band at 1576.1 ± 1.8 cm⁻¹, with $I_D/I_G = 1.24 \pm 0.08$. Both bands are broad (D: 46 cm⁻¹; G: 68 ± 10 cm⁻¹), a strong D' shoulder (21% of G) is resolved, and the 2D band is weak (2675 ± 11 cm⁻¹). These features indicate heavily defective sp² carbon. A single-wall nanotube sample normally shows a narrow, split G band and low-frequency radial breathing modes. Neither could be identified here: the sharp line at 323 cm⁻¹ and the band near 1054 cm⁻¹ also appear in the spectra of unrelated samples (DLC-II and DLC-III, Fig. 10), so they cannot be attributed to the tubes. The data therefore do not confirm the single-wall label.

The four runs differ only in integration time and agree after conversion to counts per second (run-to-run deviation 1–2 cm⁻¹ in position). A fifth run at 10 000 ms is blank: its D band has 16 counts, against about 700 expected from the other runs, so it was excluded.

4.1.4 Diamond

Diamond (Fig. 9) gives a single sp³ line at 1331.5 ± 1.0 cm⁻¹, consistent with the 1332 cm⁻¹ zone-centre phonon [4]. No D or G band is present. The line sits on a strong background that rises towards higher shifts, the signature of photoluminescence from defect centres. The peak is saturated: its top reached 98.9–99.4% of the 16-bit detector full scale in all three runs. The position was therefore fitted to the unsaturated flanks only (points below 97% of the maximum), and no linewidth is reported. A shorter integration time, as the manual advises [1], would have avoided the clipping.

4.1.5 Peak Summary

Table II collects the sp² results. All positions lie within about 25 cm⁻¹ of the nominal values for visible excitation [5]. The D/G ratio and the G width give the same ordering of disorder: HOPG, then graphene, carbon rod and nanotubes. The D band in the carbon rod and the nanotubes differs by 8 cm⁻¹, more than the combined uncertainty. Because both were excited at 532 nm, this difference reflects the samples rather than the dispersion of the D band with wavelength. Among the 2D positions, the single-peak 2D bands of the carbon rod, graphene and nanotubes cluster at 2675–2688 cm⁻¹, while only HOPG shows the $2D_2$ component of AB stacking at 2720 cm⁻¹.

4.1.6 Calibration and Resolution

Reference lines fix the calibration within 1 cm⁻¹. The unknown wafer's silicon line (Sec. 4.4) lies at 520.9 cm⁻¹, the silicon substrate under the graphene at 520.0 cm⁻¹ and diamond at 1331.5 cm⁻¹. These lie 0.7, 0.2 and 0.5 cm⁻¹ from the literature values of 520.2 cm⁻¹ [7] and 1332 cm⁻¹ [4].

The silicon line also measures the resolution. Its natural FWHM is 4.6 cm⁻¹ [7], but it appears with a FWHM of 16.6 cm⁻¹ in the unsmoothed graphene spectra. The instrument response is therefore 12 cm⁻¹ (16.6 – 4.6, if both profiles were Lorentzian) to 16 cm⁻¹ ($\sqrt{16.6^2 - 4.6^2}$, if both were Gaussian). Any band narrower than about 15 cm⁻¹ is broadened appreciably, which applies to the HOPG G band. Applying the software's smoothing filter to the unsmoothed graphene spectra widened the silicon line by 1.7 cm⁻¹, the G band by 0.5 cm⁻¹ and the 2D band by 0.1 cm⁻¹. Smoothing therefore matters only for the narrowest lines. The shoulder at 1620 cm⁻¹ is not an instrumental artefact: at 30–50 cm⁻¹ from the silicon line, the residual of the

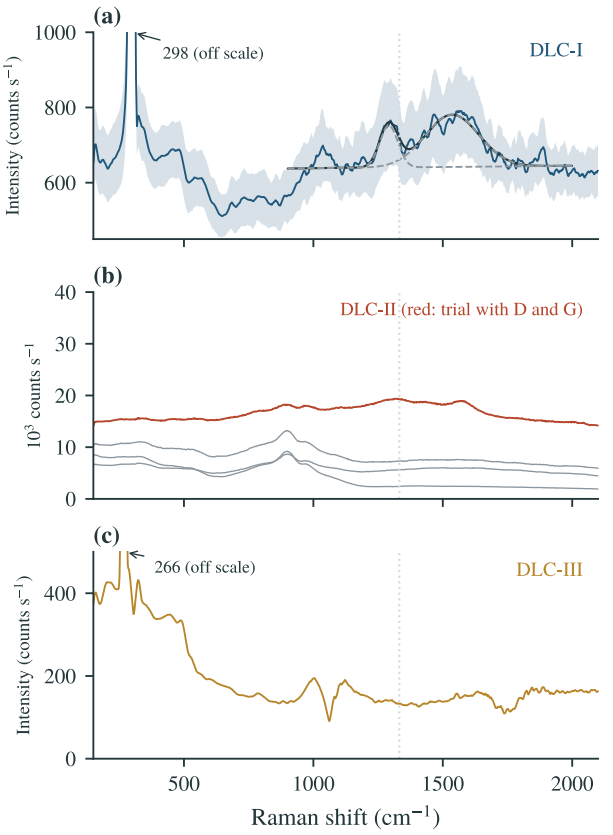


FIG. 10. Diamond-like samples in counts per second; the dotted line marks 1332 cm⁻¹. (a) DLC-I: mean of 4 runs with the run-to-run standard deviation (shaded; the runs fall into two intensity levels), the two-Gaussian D + G fit (black) and its components (dashed). (b) DLC-II: the four trials; only the red trial shows D and G bands. (c) DLC-III: mean of 5 runs from transparent and coloured regions. Sharp peaks at 298 and 266 cm⁻¹ run off scale.

silicon fit is below 2% of the peak height, against 5–21% for the shoulders above.

4.2 Diamond-like Samples

None of the three diamond-like samples is crystalline diamond: none shows a line at 1332 cm⁻¹ above the noise (RMS 19, 33 and 2 counts s⁻¹ for DLC-I to III), whereas the diamond reference gave its 1332 cm⁻¹ line on a stronger background. The three spectra otherwise differ (Fig. 10).

DLC-I (Fig. 10 a) shows the broad bands of amorphous carbon. A two-Gaussian fit on a linear background gives D at 1291 ± 5 cm⁻¹ (FWHM ≈ 80 cm⁻¹), G at 1536 ± 10 cm⁻¹ (FWHM ≈ 250 cm⁻¹) and $I_D/I_G = 0.81 \pm 0.02$ (run-to-run standard deviations; narrowing the fit range to 1000–1900 cm⁻¹ changes these by less than 3 cm⁻¹ and 0.01). The bands are only about six times the noise, so the decomposition into two Gaussians is model-dependent. A G band near 1540 cm⁻¹ with such widths places the film in stage 2 of the Ferrari–Robertson model, amorphous carbon [6]. The strongest feature of the spectrum, however, is a sharp line at 298 cm⁻¹ (2700 counts s⁻¹), which no carbon phase produces. It lies near the 300.7 cm⁻¹ phonon of crystalline germanium [7], a common substrate for DLC coatings, but the substrate was not identified.

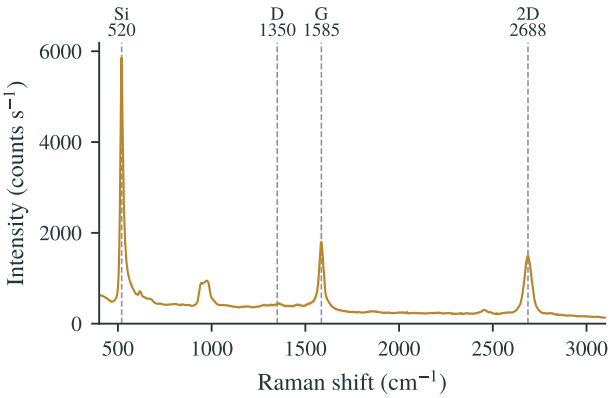


FIG. 11. Graphene on SiO_2/Si: mean of 5 locations, drawn as in Fig. 6. The silicon substrate line is at 520 cm⁻¹; the D band is weak.

DLC-II (Fig. 10 b) varies strongly between trials, which were probably taken at different spots. One trial shows a D band near 1350 cm⁻¹ and a G band near 1575 cm⁻¹ ($I_D/I_G \approx 1.1$) on a background of about 1.5×10^4 counts s⁻¹. That background is consistent with the photoluminescence of hydrogenated a-C:H [6], although its hydrogen content was not determined. In the other three trials a broad band near 900 cm⁻¹ and a weaker continuum dominate, and the D + G fit does not converge to physical values.

DLC-III (Fig. 10 c) shows no D or G band: the fitted G height is below the residual noise. Its continuum (162 counts s⁻¹ at 1800–2000 cm⁻¹) is four times weaker than that of DLC-I (645 counts s⁻¹), so photoluminescence does not hide a carbon signal. The spectrum is instead dominated by sharp lines at 266 and 323 cm⁻¹ and a band near 1050 cm⁻¹. Its five runs combine a transparent and a coloured region of the sample. The data give no evidence that DLC-III contains amorphous carbon.

4.3 Graphene Thickness

The graphene film (Fig. 11) is not a monolayer. Its 2D band is a single symmetric peak at 2687.7 ± 1.5 cm⁻¹, but its FWHM of 47 ± 4 cm⁻¹ is much broader than the about 30 cm⁻¹ that a monolayer line (about 24 cm⁻¹ intrinsically) would show with this instrument's 12–16 cm⁻¹ response [5]. The 2D/G height ratio, 1.0 ± 0.5 over the five locations (0.42 to 1.59), is also below the monolayer value of more than 1 [5]. The two measures vary together: the location with the lowest ratio has the broadest 2D band (52 cm⁻¹), and the two with the highest ratios have the narrowest (42 and 45 cm⁻¹). This pattern is consistent with thickness varying across the film between bilayer and few-layer graphene. A single broad 2D peak can also arise from rotationally misaligned layers, so these data do not fix the number of layers.

The G band lies at 1585 ± 2 cm⁻¹ with a FWHM of 27 ± 2 cm⁻¹. The 2 cm⁻¹ scatter in G position between locations, larger than in the other samples, is consistent with varying doping or strain. The D band near 1350 cm⁻¹ is weak ($I_D/I_G = 0.04 \pm 0.02$; the uncertainty is dominated by the baseline choice). It could come from flake edges within the laser spot or from point defects, and the data cannot separate the two. The silicon line confirms that the laser reached the substrate through the film and the oxide.

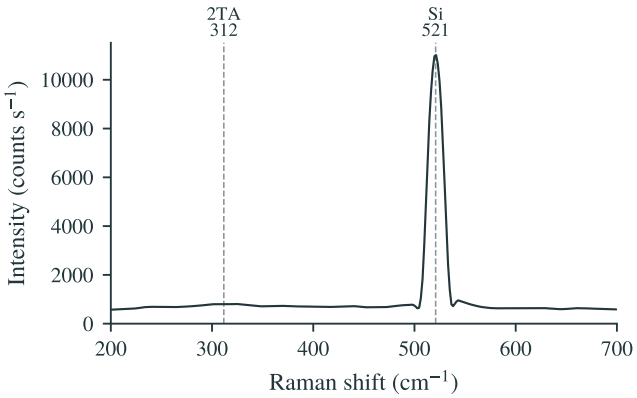


FIG. 12. Unknown wafer: mean of 5 runs, drawn as in Fig. 6. The 521 cm^{-1} line identifies crystalline silicon; the weak broad band near 312 cm^{-1} is the second-order 2TA band.

The graphene signal was strong for a film a few atoms thick. The usual explanation, which the manual asks about [1], is optical interference in the SiO_2 layer, which enhances both the exciting and the scattered fields at the graphene. This enhancement was not tested here. Comparing graphene and graphite intensities would require equal laser power, which was not recorded.

4.4 Unknown Semiconductor

The unknown wafer (Fig. 12) is crystalline silicon. Its only strong line lies at 520.9 ± 1.0 cm^{-1} (FWHM 15.9 cm^{-1}, instrument-limited), 0.7 cm^{-1} from the silicon optical phonon at 520.2 cm^{-1} [7]. A broad band at 312 cm^{-1} (FWHM 46 cm^{-1}, 1% of the main line) is consistent with silicon's second-order two-phonon (2TA) band. It is too broad to be the instrument-limited GaAs phonon near 292 cm^{-1}, and no line of GaN or other common semiconductors appears.

5. DISCUSSION

The same element gives entirely different spectra depending on its bonding. Diamond shows one sp³ line at 1331.5 cm^{-1}. The four sp² samples share the G band near 1580 cm^{-1} but differ in their defect and stacking signatures. HOPG has no detectable D band and a two-component 2D band from AB stacking. The carbon rod has a moderate D band and a single, turbostratic 2D band. The nanotube sample has a D band stronger than G. Graphene lies between HOPG and the carbon rod in disorder, and is thinner than bulk graphite by its 2D shape. This answers the manual's question of why the allotropes differ: the Raman spectrum probes the bonding and long-range order, not the composition.

The D/G ratio and the G width rank the samples identically, but the two measures have different limits. The D/G ratio is robust for the nanotubes (1.24 ± 0.08). For the weak D bands it depends on the baseline: changing from a linear to a quadratic background moves the graphene ratio from 0.05 to 0.02 and the carbon-rod ratio from 0.15 to 0.11. Only the ordering, not the absolute values, is secure at this level. The G width is limited instead by the 12–16 cm^{-1} instrument response, so the HOPG width of 18 cm^{-1} is mostly instrumental and is an upper bound on the intrinsic value.

Repeat measurements served two purposes, which answers the manual's first question [1]. Repeats on one spot (carbon rod, HOPG, diamond, wafer) agreed within 0.2–2 cm^{-1} in position and a few percent in height. This bounds the measurement noise and showed that peak positions are limited by calibration (1 cm^{-1}), not by noise. Repeats at different spots (graphene) or different trials (DLC-II) revealed real sample variation: the graphene 2D/G ratio varied by a factor of four, and only one of four DLC-II trials showed carbon bands. The repeats also exposed two acquisition failures: the blank 10 s nanotube run and the saturated diamond line. Either would have biased an average had it gone unnoticed.

The diamond-like assessment illustrates why DLC must be judged by its sp² bands. DLC-I has the broad D and G bands of amorphous carbon and no 1332 cm^{-1} line: it is an amorphous carbon film, not diamond. DLC-II shows D and G bands on a strong background consistent with a-C:H in one trial only. DLC-III shows no carbon bands at all, so the data neither identify it as DLC nor as diamond. The sharp lines at 266, 298 and 323 cm^{-1} come from materials other than carbon that these spectra do not identify.

Two further limitations affect the interpretation. First, the laser power at the sample was not recorded, so absolute intensities cannot be compared between samples; this prevents a test of the graphene enhancement and of laser heating as the cause of the carbon-rod G shift. Ratios within a spectrum are unaffected. Second, the nanotube and DLC averages combine runs at different integration times, and in DLC-III two regions of the sample. Normalising to counts per second makes this valid for peak positions and ratios, but the DLC-I runs still fall into two intensity levels, probably from refocusing.

6. CONCLUSION

The sp² allotropes were distinguished by their D, G and 2D bands. HOPG showed no D band ($I_D/I_G < 0.02$), a G band at 1581 ± 1 cm^{-1} and the two-component 2D band of AB-stacked graphite. The carbon rod had $I_D/I_G = 0.13 \pm 0.02$ and a single 2D band. The nanotubes had $I_D/I_G = 1.24 \pm 0.08$, consistent with heavily defective tubes, though the single-wall label was not confirmed. Diamond gave its sp³ line at 1331.5 ± 1.0 cm^{-1}. None of the three diamond-like samples is crystalline diamond. DLC-I is amorphous carbon (D at 1291 ± 5 cm^{-1}, G at 1536 ± 10 cm^{-1}, $I_D/I_G = 0.81 \pm 0.02$), DLC-II showed D and G bands on a strong photoluminescence background in one trial, and DLC-III showed no carbon bands. The graphene film is thicker than a monolayer (2D FWHM 47 ± 4 cm^{-1}, 2D/G 1.0 ± 0.5), but its layer number was not determined. The unknown wafer is crystalline silicon (520.9 ± 1.0 cm^{-1}).

Specific improvements follow from the limitations found. Measuring the power at the objective for each sample would allow intensity comparisons and a test of laser heating: repeating the carbon-rod spectrum at two powers would show whether its 15 cm^{-1} G downshift is thermal. Keeping the diamond peak below about 80% of full scale by shortening the integration time would make its width measurable. Recording the composition

and substrate of each diamond-like sample would allow the sharp non-carbon lines to be assigned. Mapping several points across the graphene flake, together with the 100× objective for a smaller spot, would separate thickness variation from edge contributions to the D band. Disabling smoothing would remove the 1.7 cm^{-1} broadening of the narrowest lines.

7. REFERENCES

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